

BUSINESS UNDERSTANDING REPORT

AnalystLab Africa Consulting has been engaged to prepare a clinical dataset for a future machine-learning project aimed at predicting the presence of heart disease.

The selected dataset is the Heart Failure Prediction Dataset available on Kaggle. It contains 918 patient observations and 12 variables, including 11 clinical predictor variables and one target variable, HeartDisease. The dataset includes variables such as age, sex, chest-pain type, resting blood pressure, cholesterol, maximum heart rate, exercise angina, fasting BS, and ST-slope characteristics.

The objective of this project is not to diagnose patients. Instead, the objective is to prepare the dataset for a future machine-learning classification model. The raw data must first be inspected, cleaned, transformed, encoded, scaled, and evaluated for potentially problematic observations.

1. Business Problem

The business problem is to determine how clinical patient information can be transformed into a reliable machine-learning-ready dataset for predicting whether a patient has heart disease.

Poor-quality data can negatively affect the performance and reliability of machine-learning models. Missing values, duplicate records, inconsistent data types, categorical variables, different numerical scales, outliers, and irrelevant features must therefore be addressed before modelling.

2. Project Objective

The main objective is: To clean, transform, engineer, and select features from the Heart Failure Prediction Dataset to produce a high-quality machine-learning-ready dataset for predicting the presence of heart disease.

The specific objectives are to:

- Inspect the structure and quality of the dataset.
- Identify missing values and duplicate records.
- Validate data types.
- Examine numerical and categorical variables.
- Detect potential outliers.
- Perform appropriate feature engineering.
- Encode categorical variables.
- Scale numerical variables.
- Investigate feature correlations.
- Identify relevant and potentially redundant features.
- Produce a final machine-learning-ready dataset.
- Document all preprocessing decisions.

3. Target Variable

The target variable is HeartDisease.

- 0 = No heart disease
- 1 = Heart disease

The resulting machine-learning problem is therefore a binary classification problem.

4. Importance of Predictive Modelling

Predictive modelling can help identify patterns in historical clinical data that are associated with heart disease. A future machine-learning model could potentially support healthcare professionals by providing an additional analytical tool for identifying patients who may require further clinical assessment.

However, a machine-learning model should not be interpreted as a replacement for professional medical diagnosis.

5. Expected Business Impact

A well-prepared dataset can provide a reliable foundation for future predictive modelling.

Improved data quality can contribute to:

- More reliable machine-learning models.
- Better understanding of important clinical features.
- More consistent data-processing procedures.
- Reduced risk of data-quality problems.
- Faster development of future predictive models.
- Better analytical support for healthcare decision-making.